Online Course Enrollment - Interview Problem

Business Case

An online learning company wants to investigate why course enrollment revenue is lower than expected. The business team suspects that the issue may be due to failed payments, incomplete course enrollments, discount usage that did not convert into revenue, city-level issues, course-category issues, or messy data quality problems.

You are given the role of a data analyst. Create a messy enrollment dataset, clean it, investigate it using Pandas, and present business findings from the output.

Dataset Requirement

Create an online course enrollment dataset with at least 15 records.

The dataset must contain the following columns:

- enrollment_id
- student_name
- city
- course_name
- course_category
- course_fee
- payment_status
- completion_status
- discount_code

Messy Data Requirement

The dataset should intentionally include realistic data issues commonly seen in business datasets and data analyst interviews.

Include at least the following issues:

- Extra spaces in city names.
- Different letter cases in city names or course categories.
- Course fees with currency symbols or text format, such as Rs. 5000 or INR 5000.
- At least one duplicate enrollment record.
- At least one missing student name.
- At least two failed payments.
- At least two incomplete course enrollments.
- Blank or missing discount codes.
- At least one high-value failed enrollment above INR 5000.

Cleaning Requirement

Before analysis, clean the dataset in a way that is suitable for business reporting.

Your cleaning work should include:

- Standardize city names.
- Standardize course category values.
- Clean course_fee and convert it into a numeric column.
- Handle missing student names appropriately.
- Handle blank or missing discount codes appropriately.
- Identify duplicate enrollment records and remove exact duplicates.
- Make sure the cleaned data can be used for revenue and filtering analysis.

Business Logic

Create two revenue-related columns:

- gross_fee: the raw course fee value before applying business rules.
- valid_fee_collected: the fee that should be counted as actual collected revenue.

For this problem, valid_fee_collected should be counted only when payment_status is Paid and completion_status is Completed. If payment failed or the course is incomplete, valid_fee_collected should be 0.

Interview-Style Tasks

1. Create gross_fee.
2. Create valid_fee_collected using the business rule.
3. Filter failed payment enrollments.
4. Filter incomplete course enrollments.
5. Filter paid but incomplete enrollments.
6. Filter enrollments where payment failed OR course is incomplete.
7. Filter high-value failed enrollments above INR 5000.
8. Filter enrollments from Chennai, Bangalore, and Coimbatore.
9. Filter course fees between INR 3000 and INR 10000.
10. Filter discount-code enrollments.
11. Filter discount-code enrollments with zero valid fee collected.
12. Find course-category issue count.
13. Find city issue count.
14. Write 5 business insights from the filtered outputs.

Expected Analysis Focus

The goal is not only to write Pandas code. The goal is to explain what the filtered results mean for the business.

For every filter, add a short business explanation. Example: failed payment enrollments represent potential revenue leakage; paid but incomplete enrollments may indicate course engagement or customer success issues; discount-code enrollments with zero valid fee collected may indicate poor campaign conversion.

Submission Requirement

- Submit the Python/Pandas code.
- Show the output for each task.
- Write business meaning for every filtered output.
- Write 5 final business insights.
- Do not submit only code without explanation.

Final Deliverable

The final answer should clearly show how a data analyst would investigate a messy course-enrollment dataset, clean it, apply business rules, filter problem records, and explain the business meaning of the findings.